Kshitiz Tyagi

Indian Institute of Technology Kanpur
Biological Sciences & Bioengineering
Final Year Undergraduate

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Course	College / University	Year	CGPA / %
Bachelor of Technology (B.Tech)	Indian Institute of Technology Kanpur	2023 - Present	8.8/10.0
CBSE (Class XII)	Dewan Public School, Hapur	2022	95.4%
ICSE (Class X)	S.D.A Senior Secondary School, Hapur	2020	89.0%

Scholastic Achievements

- Secured **Academic Excellence Award (2024–2025)** for placing within the top 10% of the batch with consecutive 9.4 SPIs.
- Selected as a **Reliance Foundation Scholar (2023–2024)** through a competitive nationwide test among 58,000 applicants.

Work Experience

Battery Smart
May 2026 – Jul 2026

Data Science Intern | Project: Second Brain

- Engineered event-driven feedback pipeline for self-improving RAG system using **MCP** to auto-log feedback, orchestrate expert review, and ship validated knowledge-base corrections in **3–5s** via GitHub **REST API** and **PostgreSQL**, with atomic SQL operations preventing multi-pod race conditions across replicas.
- Architected a 6-stage NLP pipeline to extract structured **temporal triplets** from conversational transcripts via a headless LLM (**Kiro CLI**); processed a **1,613**-conversation backlog at **$0 inference cost** with a **MiniLM** PII filter under a **1GiB** RAM budget.
- Drove domain adaptation by authoring and verifying ∼**6,200 lines** of schema documentation across **230+ Databricks tables**; developed a 12-page **SQL RCA playbook** to systematically diagnose and reduce LLM-generated query errors.
- Developed a hybrid semantic retrieval router combining **MiniLM embeddings**, **BM25**, and keyword filters using **RRF**; mapped cross-domain dependencies across a **1,248-node / 92-community Graphify** graph to eliminate routing blind spots.
- Diagnosed representation bias in a production **YOLO** face-verification model related to facial hair; relabeled data in **CVAT** and fine-tuned for 50 epochs across ∼**95,000 images**, detecting **3.3x more defects** and improving confidence from **0.84 to 0.96**.

Skaeo
May 2025 – Jul 2025

ML and AI Systems Intern

- Engineered an edge Face Anti-Spoofing pipeline for **My RollCall Pro** using **dlib (HOG + 68 landmarks)** to track blink (EAR), smile (MAR), and pupil gaze, achieving ∼**5s latency** against unauthorized 2D biometric spoof presentation attacks.
- Benchmarked multiple OCR engines and customized **RapidOCR with ONNX Runtime** for serial number recognition; developed end-to-end preprocessing, extraction and validation logic, reaching ∼**2s inference** in production deployment pipelines.
- Built an agentic automation pipeline using **LangChain**, **Phidata**, and a **Telegram bot** for URL parsing, SWOT analysis, and scheduled posting, streamlining marketing workflows and cutting manual turn around time by 40 %.

Publications

An Efficiency Hierarchy Model for Perceptual Grouping
Peer-reviewed Poster at CogSci 2026 — *Published - 2026*

- Built a computational model to statistically analyze human reaction times and visual grouping performance across test trials.

Plural Pathways to Computational Thinking
Co-author; submitted to RPTEL — *Under-Review*

- Performed regression analysis and hypothesis testing to evaluate how student problem-solving strategies differ across age groups.

Key Projects

Adaptive Spatial Learning Platform
Aug 2026 - Ongoing

Prof. Anveshna Srivastava | Undergraduate Project

- Building a full-stack adaptive learning platform using **React**, **FastAPI**, and **PostgreSQL** to manage sessions and structured user data.
- Implementing a real-time feedback engine alongside interactive **3D LEGO modules** with auto task validation and adaptive prompting.
- Engineering scalable **AWS architecture** supporting new learning modules, async event logging pipelines, and structured user analytics.

CineScope: Visual Analytics
May 2025 – Jul 2025

Prof. Soumya Dutta | CS 661 Course Project

- Created a dynamic **Dash & Plotly** analytics dashboard featuring real-time cross-filters, streamgraphs, and choropleth maps.
- Preprocessed and clustered metrics across **900K+ titles** using **PCA and t-SNE**, designing **7+ interactive multivariate visualizations** to analyze box-office trends, genre popularity, and studio ROI.

SynapticSpike: Statistical Modeling
Jan 2025 – May 2025

Prof. Arjun Ramakrishnan | BSE 662 Project

- Implemented 3-level Bayesian **Hierarchical Gaussian Filters** across **33 subjects** to model trial-by-trial belief updates and volatility.
- Scaled mixed-effects regressions (**LMER**) over **1.7M+ observations** on an **HPC cluster**, modeling longitudinal variance and random effects.

Word For Thought & The Minds Mirror
May 2024 – Jan 2025

Brain & Cognitive Science Club | Project Mentor

- Led a 20+ member team developing a speech-emotion pipeline, converting raw audio into spectrograms via **FFT** and training a **2D CNN** to achieve **97% accuracy**.
- Curated a 200-sample visual-stimulus EEG dataset using **PsychoPy**; extracted spectral features using **FFT**, **STFT**, **Autocorrelation**, and **Welch's method** with temporal noise filtering.
- Trained a **1D CNN** with **SMOTE** to achieve **90% accuracy** on visual trials, benchmarking against a **BiLSTM** for multiclass classification.

BioBytes: Genomic Classifier
May 2024 – Jun 2024

BioSoc | IIT Kanpur

- Built a robust DNA classifier using k**-mer features** and machine learning models like **Naive Bayes**, **Random Forest** and **XGBoost**.
- Handled zero-frequency issues via **Laplace Smoothing** and utilized **CountVectorizer**; addressed class imbalance and feature dimensionality.

Technical Skills

- **Languages:** Python, C++, C, Java, SQL, R
- **ML/AI:** PyTorch, TensorFlow, LangChain, OpenCV, YOLO, dlib, ONNX Runtime, XGBoost, Librosa, Hugging Face
- **Data/Backend:** Pandas, NumPy, Seaborn, FastAPI, PostgreSQL, Redis, Databricks, React
- **Tools/Cloud:** Git, Docker, AWS, MCP, CUDA, PsychoPy, Bash, LaTeX

Extracurricular Activities

- **Overall Coordinator, BioSoc:** Directed 6-vertical core team and utilized an **INR 2 Lakh budget**, driving initiatives, seminars, and mentorship.
- **Secretary, BCS Club:** Mentored **40+ members** and hosted GPU/ML workshops with top NVIDIA Inception startups for **200+ students**.
- **Galaxy '25 (Inter-Hall Cultural Festival):** Won the **Gold Medal** in Group Dance, delivering thematic synchronized stage performances.

Relevant Courses

(* Online Certification)

Introduction to Machine Learning	Data Structures & Algorithms	Fundamentals of Computing	Big Data Visual Analytics
Applied Probability & Statistics	Neural Networks & Deep Learning*	Convolutional Neural Networks*	Introduction to Electronics